

UAV Swing Payload Model

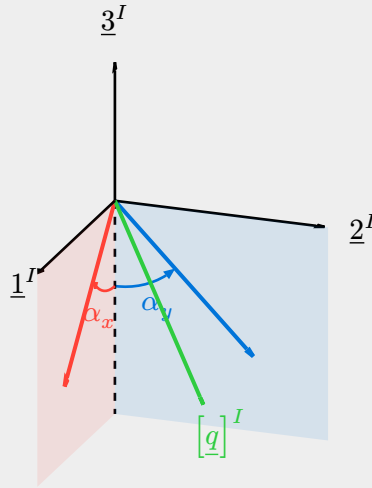
$$m_P \underline{a}_P = T \underline{q} - m_P \underline{g} \quad (1)$$

$$m_D \underline{a}_D = \underline{f} - T \underline{q} - m_D \underline{g} \quad (2)$$

$$\frac{d}{dt} \omega^{BE} = (J_B^B)^{-1} [\underline{n}_B - \underline{\Omega}^{BE} J_B^B \omega^{BE}] \quad (3)$$

$$\underline{s}_P = \underline{s}_D - L \underline{q} \quad (4)$$

Swing angles



$$[\underline{q}]^I = \begin{pmatrix} \cos \alpha_x & 0 & -\sin \alpha_x \\ 0 & 1 & 0 \\ \sin \alpha_x & 0 & \cos \alpha_x \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha_y & -\sin \alpha_y \\ 0 & \sin \alpha_y & \cos \alpha_y \end{pmatrix} [-\underline{3}^I]^I = \begin{pmatrix} \sin \alpha_x \cos \alpha_y \\ \sin \alpha_y \\ -\cos \alpha_x \cos \alpha_y \end{pmatrix} \approx \begin{pmatrix} \alpha_x \\ \alpha_y \\ -1 \end{pmatrix} \quad (5)$$

ArUco Marker Tracking EKF

Frames

$[\cdot]^I$: ENU (+x East, +y North, +z Up)

$[\cdot]^B$: body (+x nose, +y portside, +z up)

$[\cdot]^C$: camera (+x right, +y up, +z optical axis)

(6)

State

$$[\xi]^I = \begin{pmatrix} \alpha_x \\ \alpha_y \\ \dot{\alpha}_x \\ \dot{\alpha}_y \\ \psi_P \end{pmatrix} \in \mathbb{R}^5 \quad (7)$$

$$s_x = \sin \alpha_x, \quad c_x = \cos \alpha_x, \quad s_y = \sin \alpha_y, \quad c_y = \cos \alpha_y \quad (8)$$

α_x, α_y : tether swing angles, tipping toward +x^I (East) and +y^I (North) [rad]

$\dot{\alpha}_x, \dot{\alpha}_y$: swing rates [rad/s]

ψ_P : payload yaw: marker-row direction [rad]

Input

$$[T]^{IB} = \begin{pmatrix} c_\theta c_\psi & s_\varphi s_\theta c_\psi - c_\varphi s_\psi & c_\varphi s_\theta c_\psi + s_\varphi s_\psi \\ c_\theta s_\psi & s_\varphi s_\theta s_\psi + c_\varphi c_\psi & c_\varphi s_\theta s_\psi - s_\varphi c_\psi \\ -s_\theta & s_\varphi c_\theta & c_\varphi c_\theta \end{pmatrix} \quad (9)$$

Euler angles: φ : roll θ : pitch ψ : yaw

$$[\underline{a}_D]^I = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = [T]^{IB} [\underline{a}_D]^B \quad (10)$$

$[\underline{a}_D]^I$: acceleration inputs from the drone [m/s²]

Process model

$$\begin{bmatrix} \dot{\xi} \end{bmatrix}^I = f(\xi, a) = \begin{pmatrix} \dot{\alpha}_x \\ \dot{\alpha}_y \\ -\frac{c_x a_1 + s_x a_3}{L c_y} + 2 \frac{s_y}{c_y} \dot{\alpha}_x \dot{\alpha}_y \\ -\frac{c_y a_2 + s_y (c_x a_3 - s_x a_1)}{L} - s_y c_y \dot{\alpha}_x^2 \\ 0 \end{pmatrix} \approx \begin{pmatrix} \dot{\alpha}_x \\ \dot{\alpha}_y \\ -\frac{1}{L} (a_1 + \alpha_x \|g\|) \\ -\frac{1}{L} (a_2 + \alpha_y \|g\|) \\ 0 \end{pmatrix} \quad (11)$$

L : drone tether-pivot to payload CG [m]

$$F = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ -\frac{\|g\|}{L} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\|g\|}{L} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad (12)$$

$F \in \mathbb{R}^{5 \times 5}$: process Jacobian (continuous-time)

Prediction

Jacobian to discrete-time conversion w/ Euler integration.

$$\hat{\xi}_{k+1}^- = \hat{\xi}_k^+ + \Delta t f(\hat{\xi}_k^+, a_s) \quad (13)$$

$$\Phi_k = I_5 + F \Delta t, \quad Q = \text{diag}(0, 0, q_x, q_y, q_\psi) \Delta t \quad (14)$$

$$P_{k+1}^- = \Phi_k P_k^+ \Phi_k^\top + Q \quad (15)$$

Δt : time step [s]

$\hat{\xi}_k^-, \hat{\xi}_k^+$: estimate before & after the update at step k

$P_k^-, P_k^+ \in \mathbb{R}^{5 \times 5}$: covariance before & after

Φ_k : discrete time Jacobian

q : process-noise intensity

Shelving Jacobian to discrete-time conversion w/ RK4 for now.

$$k_1 = f(\hat{\xi}_k^+, a_s), \quad k_2 = f\left(\hat{\xi}_k^+ + \frac{\Delta t}{2} k_1, a_s\right) \quad (16)$$

$$k_3 = f\left(\hat{\xi}_k^+ + \frac{\Delta t}{2} k_2, a_s\right), \quad k_4 = f(\hat{\xi}_k^+ + \Delta t k_3, a_s)$$

$$\hat{\xi}_{k+1}^- = \hat{\xi}_k^+ + \frac{\Delta t}{6} (k_1 + 2k_2 + 2k_3 + k_4) \quad (17)$$

$$\Phi_k = I_5 + F \Delta t + \frac{1}{2} (F \Delta t)^2, \quad Q = \text{diag}(0, 0, q, q, q_\psi) \Delta t \quad (18)$$

$$P_{k+1}^- = \Phi_k P_k^+ \Phi_k^\top + Q \quad (19)$$

Δt : time step [s]

$\hat{\xi}_k^-, \hat{\xi}_k^+$: estimate before & after the update at step k

$P_k^-, P_k^+ \in \mathbb{R}^{5 \times 5}$: covariance before & after

Φ_k : discrete time Jacobian

q : process-noise intensity

Measurement

Pre-processing logic

For every $j \in V_k$, solvePnP returns the marker pose in the camera frame,

$$[t_j]^C, [T]^{CM_j} \quad j \in V_k \quad (20)$$

V_k : marker IDs detected in frame k

$[t_j]^C$: marker j center in the camera frame [m]

$[T]^{CM_j}$: transformation to the camera frame from the marker local frame M_j

The first column of $[T]^{CM_j}$ is the marker x -axis, which runs along the marker row,

$$[m_j]^C = [T]^{CM_j} [\underline{1}^{M_j}]^{M_j} \quad (21)$$

Each marker gives its own estimate of the board center; average over the markers actually detected in the frame,

$$[p_{\text{ctr}}]^C = \frac{1}{m_k} \sum_{j \in V_k} \left([t_j]^C + o_j [m_j]^C \right) \quad (22)$$

$[p_{\text{ctr}}]^C$: payload center in the camera frame

m_k : # of markers in the frame

o_j : marker offset from board center along the row, +left [m]

Shift the origin from the camera optical center to the tether pivot,

$$[p]^C = [p_{\text{ctr}}]^C + [T]^{CB} ([t_{BC}]^B - [\ell]^B) \quad (23)$$

$[p]^C$: payload center after the camera offset is corrected, camera frame [m]

$[t_{BC}]^B$: camera optical center in the body frame [m]

$[\ell]^B$: drone tether pivot in the body frame [m]

Average the payload yaw and transform into the inertial frame,

$$[m]^I = [T]^{IB} [T]^{BC} \left(\frac{1}{m_k} \sum_{j \in V_k} [m_j]^C \right) \quad (24)$$

$$z_{\psi_P} = \text{atan2}(m_2^I, m_1^I)$$

Measurement model

After some pre-processing from pixels to camera frame coordinates, we measure,

$$z_k = \begin{pmatrix} \frac{p_x^C}{\|p\|} \\ \frac{p_y^C}{\|p\|} \\ z_{\psi_P} \end{pmatrix} + \varepsilon_k \quad (25)$$

p : payload center position

z_{ψ_P} : measured payload yaw

ε_k : measurement noise

Prediction of measurement

Measurement predicted from swing angles,

$$h(\xi) = \begin{pmatrix} q_x^C \\ q_y^C \\ \psi_P \end{pmatrix} \quad [\underline{q}]^C = [T]^{CB} [T]^{BI} \begin{pmatrix} \alpha_x \\ \alpha_y \\ -1 \end{pmatrix} \quad (26)$$

$$H = \frac{\partial h}{\partial \xi} = \begin{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} [T]^{CB} [T]^{BI} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \\ (0 \ 0 \ 0 \ 0 \ 1) \end{pmatrix} \quad (27)$$

Update

Innovation

$$y_k = z_k - h_k(\hat{\xi}_k^-) \quad (28)$$

$$R_k = \text{diag}(\sigma_x^2, \sigma_y^2, \sigma_\psi^2) \in \mathbb{R}^{3 \times 3} \quad (29)$$

$$S_k = H_k P_k^- H_k^\top + R_k \in \mathbb{R}^{3 \times 3} \quad (30)$$

y_k : innovation
 R_k : measurement noise
 S_k : innovation covariance

Kalman gain

$$K_k = P_k^- H_k^\top S_k^{-1} \in \mathbb{R}^{5 \times 3} \quad (31)$$

$K_k \in \mathbb{R}^{5 \times 3}$: Kalman gain

$$\hat{\xi}_k^+ = \hat{\xi}_k^- + K_k y_k, \quad (32)$$

$$P_k^+ = (I_5 - K_k H_k) P_k^- \quad (33)$$

If no markers detected:

$$\hat{\xi}_k^+ = \hat{\xi}_k^-, \quad P_k^+ = P_k^- \quad (34)$$

Control

State-Space

$$\dot{x} = Ax + Bu \quad (35)$$

where,

$$x = \begin{pmatrix} s_1^I \\ s_2^I \\ s_3^I \\ v_1^I \\ v_2^I \\ v_3^I \\ \alpha_x^I \\ \alpha_y^I \\ \dot{\alpha}_x^I \\ \dot{\alpha}_y^I \end{pmatrix}, \quad u = \begin{pmatrix} a_1^I \\ a_2^I \\ a_3^I \end{pmatrix} \quad (36)$$

$$\text{drone states} \begin{cases} \dot{s}_1^I = v_1^I \\ \dot{s}_2^I = v_2^I \\ \dot{s}_3^I = v_3^I \\ \dot{v}_1^I = a_1^I \\ \dot{v}_2^I = a_2^I \\ \dot{v}_3^I = 0 \end{cases} \quad (37)$$

$$\text{payload states} \begin{cases} \dot{\alpha}_x^I = \dot{\alpha}_x^I \\ \dot{\alpha}_y^I = \dot{\alpha}_y^I \\ \ddot{\alpha}_x^I = -\frac{g}{L}\dot{\alpha}_x^I - \frac{1}{L}a_1^I \\ \ddot{\alpha}_y^I = -\frac{g}{L}\dot{\alpha}_y^I - \frac{1}{L}a_2^I \end{cases} \quad (38)$$

$$\Rightarrow A, B \quad (39)$$

Equilibrium

$$x^* = \begin{pmatrix} [s_{PL}(t)]^I + \begin{pmatrix} 0 \\ 0 \\ L \end{pmatrix} \\ [v_{PL}]^I \\ 0 \end{pmatrix} \quad (40)$$

LQI Controller

Control Law

$$u = -(K_x \ K_I) \begin{pmatrix} e \\ \int e \end{pmatrix} \quad (41)$$

$$e = x - x^*$$

$$y_{\text{error}} = Ce \quad (42)$$

$$\int e[k+1] = \int e[k] + y_{\text{error}} \Delta t \quad (43)$$

Output

$$y = Cx \quad (44)$$

$$\begin{aligned} y &= [s_{PL}]^I \\ &= \begin{pmatrix} s_1^I - L\alpha_x^I \\ s_2^I - L\alpha_y^I \\ s_3^I - L \end{pmatrix} \end{aligned} \quad (45)$$

$$\Rightarrow C \quad (46)$$

Gain matrix

$$\overline{K} = (K_x \ K_I) \quad (47)$$

Augmented State-space

$$\dot{z} = \overline{A}z + \overline{B}u \quad (48)$$

$$z = \begin{pmatrix} x \\ \int [s_{PL}]^I \end{pmatrix} \quad (49)$$

$$\Rightarrow \bar{A}, \bar{B} \quad (50)$$

Cost minimization

Solve for the optimal $\bar{K}$,

$$J = \int_0^\infty \left(\begin{pmatrix} e \\ \int e \end{pmatrix}^\top \bar{Q} \begin{pmatrix} e \\ \int e \end{pmatrix} + u^\top R u \right) dt \quad (51)$$

State weighting matrix,

$$\bar{Q} = \begin{pmatrix} C^\top W C & 0_{10 \times 3} \\ 0_{3 \times 10} & W_{\text{int}} \end{pmatrix} \in \mathbb{R}^{13 \times 13} \quad (52)$$

$$W = \begin{pmatrix} w_x & 0 & 0 \\ 0 & w_y & 0 \\ 0 & 0 & w_z \end{pmatrix} \quad (53)$$

$$W_{\text{int}} = \begin{pmatrix} w_{\text{int},x} & 0 & 0 \\ 0 & w_{\text{int},y} & 0 \\ 0 & 0 & w_{\text{int},z} \end{pmatrix} \quad (54)$$

Control weighting matrix,

$$R = (\text{tuning const}) \times \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (55)$$

Solving for the gain matrix,

$$\bar{A}^\top P + P \bar{A} - P \bar{B} R^{-1} \bar{B}^\top P + \bar{Q} = 0 \Rightarrow P \quad (56)$$

$$\bar{K} = R^{-1} \bar{B}^\top P \quad (57)$$